Tumor Detection across Hospitals Using Deep Learning: A Study on Camelyon17-WILDS

ELE680 — Final Project Report

University of Stavanger | Group 7

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Project Repository: https://github.com/Bneja/ELE680-Group-7

1. Introduction

A central challenge in histopathology-based tumor detection is whether a model trained on images from certain hospitals can reliably classify images collected at other hospitals. Although the underlying task remains the same, differences in tissue staining, slide preparation, and scanning procedures can change the appearance of histopathology images. A model may therefore achieve strong performance on data similar to its training set while performing less reliably on data from a different hospital.

This project investigates this problem using Camelyon17-WILDS, a benchmark for binary tumor classification in histopathology image patches collected from five hospitals [1]. The benchmark provides predefined data splits that allow performance to be examined under hospital-related distribution shift. Rather than focusing only on classification accuracy, our experiments consider how the models perform when evaluated on the official held-out test set.

The main research question is: How do model architecture and training-time data augmentation affect tumor detection performance across hospitals? To investigate this, we compared a compact convolutional neural network (SmallCNN) and ResNet18, each trained with and without data augmentation. This resulted in four experimental configurations evaluated using test accuracy and ROC-AUC.

Our approach was to establish a common comparison of the completed experiments and examine whether the more complex architecture provided an advantage over the smaller CNN. We also investigated whether augmentation was associated with better performance under the benchmark's hospital distribution shift. The results are interpreted with attention to the differences between validation and test performance, rather than assuming that a higher score alone demonstrates reliable generalization.

2. Materials and Methods

2.1 Dataset and Experimental Setup

The experiments were conducted using Camelyon17-WILDS, a histopathology dataset containing image patches collected at five hospitals [1]. The objective was binary classification: distinguishing patches containing tumor tissue (label 1) from those without tumor tissue (label 0). Hospital-specific differences in image appearance make this dataset suitable for studying model generalization under distribution shift.

Because downloading through the WILDS Python package did not work due to a broken link, the Camelyon17 archive was retrieved from the Hugging Face repository fumiyau/wilds-archive [4]. After extracting the archive files into the local dataset directory, the dataset was loaded using the WILDS Python package. The Hugging Face repository is a mirror of the WILDS archive, rather than an official WILDS repository [4].

The experimental comparison included two architectures: a compact convolutional neural network (SmallCNN) and ResNet18 [2]. Each architecture was trained under two configurations, with and without data augmentation, resulting in four models. The experiments were conducted using Python and PyTorch [3] on the University of Stavanger's Gorina4 computing cluster. The official dataset splits were retained to support consistent evaluation and comparison across the four configurations.

2.2 Training Without Data Augmentation — Benjamin's Contribution

To establish the non-augmented baseline, I trained SmallCNN and ResNet18 using the official Camelyon17-WILDS training split. This split contains patches from hospitals 0, 3, and 4. I used the OOD validation split from hospital 1 to select the checkpoint with the highest balanced accuracy. The ID validation split was evaluated for comparison, while the official test split from hospital 2 was reserved for final evaluation.

The SmallCNN used four convolutional layers with 32, 64, 128, and 192 channels. Each convolutional layer was followed by batch normalization and a ReLU activation; max pooling followed the first three blocks. Adaptive average pooling and a two-class linear layer produced the final predictions. ResNet18 [2] was created with torchvision using no pretrained weights, and its final fully connected layer was replaced with a two-class output layer. Both models were therefore trained from scratch.

For both models, preprocessing was limited to resizing patches to 96 x 96 pixels, converting them to tensors, and normalizing each RGB channel with a mean and standard deviation of 0.5. No flips, rotations, or color transformations were applied. Training used a batch size of 8 for 16 epochs, AdamW with a learning rate of 0.001 and weight decay of 0.0001, and cross-entropy loss. A random seed of 42 was set for Python, NumPy, and PyTorch. The training data were shuffled, and after each epoch the OOD validation balanced accuracy was checked to determine whether to save a new best checkpoint. The held-out test set was evaluated after checkpoint selection.

2.3 Training With Data Augmentation — Rabia's Contribution

To investigate whether data augmentation could improve tumor detection across hospitals, I trained augmented versions of both SmallCNN and ResNet18 using the Camelyon17-WILDS training split.

The data augmentation pipeline was implemented using torchvision.transforms. After resizing the training images to 96 × 96 pixels, three augmentation techniques were applied. First, horizontal and vertical flipping, each with a probability of 0.5, introduced variation in tissue orientation. Second, random rotations of 0°, 90°, 180°, or 270° exposed the models to additional tissue orientations. Third, color jitter introduced small variations in brightness, contrast, and saturation (0.1 each), representing possible differences in image appearance across hospitals. These transformations increased the diversity of training images while preserving their classification labels. Following augmentation, the images were converted to tensors and normalized using a mean and standard deviation of 0.5 for each RGB channel.

Both augmented models used a batch size of 8, AdamW optimization with a learning rate of 0.001 and weight decay of 0.0001, and cross-entropy loss. Training was configured for 16 epochs. The training data were shuffled, while validation and test images underwent only deterministic resizing and normalization, without random augmentation.

During training, performance was monitored on the in-distribution (ID) and out-of-distribution (OOD) validation splits. The augmented SmallCNN checkpoint was selected based on the highest OOD validation balanced accuracy, achieved at epoch 16. The selected checkpoints were evaluated on the official held-out test split containing 85,054 images. The resulting performance is presented in Section 3.

2.4 Evaluation and Model Comparison

The four models were evaluated on the predefined Camelyon17-WILDS splits. For the non-augmented models, OOD validation balanced accuracy was used to select the checkpoint, and ID validation performance was evaluated for comparison. For the augmented models, both ID and OOD validation performance were monitored during training, with OOD validation balanced accuracy used for checkpoint selection. The official held-out test split was reserved for final evaluation. Each of the four configurations was trained once; we did not repeat the experiments with additional random seeds.

Three evaluation metrics were considered: classification accuracy, balanced accuracy, and area under the receiver operating characteristic curve (ROC-AUC). Accuracy measures the proportion of correctly classified image patches. Balanced accuracy calculates the average recall across the tumor-positive and tumor-negative classes, giving both classes equal importance. ROC-AUC measures how well a model distinguishes between the two classes across different classification thresholds. Together, these metrics provide a broader assessment of classification performance.

The final comparison included SmallCNN and ResNet18, each trained with and without data augmentation. I compared their test results to examine differences between the two architectures and between the augmented and non-augmented configurations. For each architecture, the change in test accuracy was calculated in percentage points. The comparison was used to identify which configuration performed best on the official test set and to discuss the models' ability to generalize across hospitals.

3. Results

3.1 Performance Comparison

The four experimental configurations were evaluated on the official Camelyon17-WILDS held-out test set. Table 1 presents their classification accuracy, balanced accuracy, and ROC-AUC scores.

Table 1. Test performance of the four model configurations.

Model	Data Augmentation	Test Accuracy (%)	Test Balanced Accuracy (%)	ROC-AUC
SmallCNN	No	85.39	85.39	0.9280
SmallCNN	Yes	91.22	91.22	0.9552
ResNet18	No	61.45	61.45	0.6526
ResNet18	Yes	81.16	81.16	0.9451

The augmented SmallCNN achieved the highest test accuracy (91.22%), balanced accuracy (91.22%), and ROC-AUC (0.9552) among the four configurations. Compared with the SmallCNN trained without augmentation, its test accuracy increased by 5.83 percentage points.

For ResNet18, the augmented configuration achieved a test accuracy of 81.16%, compared with 61.45% without augmentation, representing an increase of 19.71 percentage points. Its ROC-AUC also increased from 0.6526 to 0.9451.

Test accuracy and balanced accuracy were identical for each configuration because the official test set was class-balanced. Overall, both architectures achieved higher test accuracy, balanced accuracy, and ROC-AUC in their augmented configurations. However, the augmented SmallCNN outperformed the augmented ResNet18 on all three metrics. These results indicate that the larger ResNet18 architecture did not provide a performance advantage in this experimental comparison.

3.2 Validation and Test Performance

In addition to the final test results, the augmented SmallCNN and ResNet18 models were evaluated using the in-distribution (ID) and out-of-distribution (OOD) validation sets. Table 2 summarizes their validation and test performance.

Table 2. Validation and test performance of the augmented models.

Model	ID Validation Accuracy (%)	OOD Validation Accuracy (%)	Test Accuracy (%)
SmallCNN with augmentation	98.78	91.13	91.22
ResNet18 with augmentation	98.86	88.89	81.16

Both augmented models achieved high ID validation accuracy, exceeding 98%. Their performance was lower on the OOD validation and official test sets, where hospital-related distribution shift was considered.

The augmented SmallCNN achieved an OOD validation balanced accuracy of 91.13% and a test accuracy of 91.22%. The augmented ResNet18 achieved an OOD validation balanced accuracy of 88.89% and a test accuracy of 81.16%.

Although ResNet18 achieved slightly higher ID validation accuracy, SmallCNN performed better on the OOD validation and test sets. This highlights the importance of evaluating models beyond the training distribution when assessing their ability to generalize across hospitals.

4. Discussion

The results suggest that data augmentation improved tumor classification performance under the hospital-related distribution shift represented in Camelyon17-WILDS. Both SmallCNN and ResNet18 achieved higher test accuracy, balanced accuracy, and ROC-AUC when trained with augmentation. The improvement was particularly large for ResNet18, whose test accuracy increased by 19.71 percentage points, compared with an increase of 5.83 percentage points for SmallCNN.

An important finding was that the augmented SmallCNN outperformed the augmented ResNet18, despite having a simpler architecture. This indicates that a more complex model does not necessarily achieve better generalization across hospitals. In these experiments, the combination of SmallCNN and augmentation produced the strongest overall test performance.

The high ID validation accuracy of both augmented models, compared with their lower OOD validation and test results, also highlights the challenge of distribution shift. Strong performance on data similar to the training distribution does not guarantee equally strong performance on images collected at other hospitals. The results therefore support the importance of evaluating histopathology models using held-out hospital data.

Several limitations should be considered. The experiments were conducted on one benchmark dataset using two architectures and a limited set of augmentation techniques. Each configuration was run once, so the reported differences do not show how results vary across random seeds and should be interpreted as observations from these runs. The results may also depend on training settings and checkpoint selection. Furthermore, the experiments do not establish which individual augmentation technique contributed most to the observed improvements. In future work, stain normalisation could be explored to standardise the colour appearance of histopathology images across hospitals before training, potentially reducing hospital-specific visual differences. Repeating each configuration with multiple random seeds and testing additional model configurations would help assess the consistency of the findings.

Overall, the findings suggest that data augmentation was beneficial in this experimental setting, while model complexity alone did not guarantee improved performance under hospital-related distribution shift.

5. Conclusion

This project investigated how model architecture and data augmentation affect tumor detection performance across hospitals using the Camelyon17-WILDS dataset. Four configurations were compared: SmallCNN and ResNet18, each trained with and without data augmentation.

The results showed that data augmentation improved the test performance of both architectures. The augmented SmallCNN achieved the highest test accuracy and balanced accuracy of 91.22%, with a ROC-AUC of 0.9552. Augmentation increased test accuracy by 5.83 percentage points for SmallCNN and 19.71 percentage points for ResNet18.

Although ResNet18 is a more complex architecture, it did not outperform SmallCNN in these experiments. The differences between in-distribution validation and held-out hospital performance also demonstrated the importance of evaluating models under distribution shift.

Overall, the findings suggest that data augmentation can improve tumor classification across hospitals, while increasing model complexity alone does not guarantee better generalisation.

Individual Contributions

Rabia Zakriyya: Implemented and trained SmallCNN and ResNet18 with data augmentation. Evaluated the augmented models and compared the performance of all four experimental configurations, including the effects of augmentation and model architecture.

Benjamin: Implemented and trained SmallCNN and ResNet18 without data augmentation, establishing the baseline results used for comparison.

References

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